

Project Timeline (4 Weeks)

Dataset Collection (Week 1)

- Collect 100 classic datasets (e.g., [UCI Repository](#))
- Clean and standardize formats
- Create multiple dataset variants

Goal: Build a reliable base dataset pool

Missingness Simulation (Week 1/2)

Inject:

- MCAR (random)
- MAR (depends on observed data)
- MNAR (depends on hidden values)
- Generate ~100 datasets per type

Prompt & Agent Testing (Week 2/3)

Test under 4 conditions:

- Correct assumption
- No guidance
- Context-based reasoning
- Incorrect assumption

Goal: See if agent questions assumptions or follows blindly

Evaluation (Week 3)

Quantitative:

- MSE
- Distribution similarity

Qualitative:

- Code
- Explanations

- Assumption awareness

Output: 3-Level Trust Classification

Writing (Week 4)

- Analyze patterns
- Connect to literature
- Discuss limitations

Final Output: Complete research paper